# Proof Bundle Report — Canonical 110 mm Crystal

**VERDICT: UNCERTAINTY_OVERLAPS_CONVENTIONAL_NODE** (engine status: UNCERTAINTY_OVERLAPS_CONVENTIONAL_NODE; the verdi
not forced — a null or conventional outcome passes).

Configurations: ideal N=7 (110.03766741071429 mm = 770.263671875/7, arithmetic) and nominal (110.0 mm). Frozen alpha-quartz constants (Bechmann 1958) throughout; CPU float64 is the numerical authority (DV4-004).

## Key numbers (this run)

- ideal first elastic modes (medium mesh): [13779.3, 13779.7, 26044.8, 31882.4] Hz
- nominal first elastic modes: [13784.0, 13784.8, 26059.0, 31894.6] Hz
- orthonormality error: ideal 1.55e-15, nominal 2.00e-15
- mass patch rel err: 8.84e-16
- cavity max rel err vs exact: 2.39e-04
- Christoffel Z-axis vs frozen v3: 0.00e+00
- scrambled-field null control: NO_STABLE_CANDIDATE

## Eye

UNCERTAINTY_OVERLAPS_CONVENTIONAL_NODE: see eye/consensus.json and docs/plans-v4/EYE_DIAGNOSTIC_REPORT_110MM.md. eye_coordinate is null.

## Provenance

See PROVENANCE.json. Acceleration results are copied from evidence/v4/agent07 (hardware-dependent); the tuning-fork V.9 CSV is copied from evidence/v4/agent10; everything else regenerated live.